# PRIME PRIMITIVE ROOTS BELOW EVERY POWER OF $p$

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Abstract. For every $\eta > 0$ every prime $p$ has a prime primitive root $\ell \ll_\eta (\tau(p-1)^2 \log p)^{8+\eta}$. Hence the least primitive root is $O_\varepsilon(p^\varepsilon)$, as Vinogradov conjectured, and every large prime has a smaller prime primitive root. The proof uses the zero-free half-plane $\operatorname{Re} s > 7/8$ and is checked in Lean 4.

## 1. Introduction

For a prime $p$ let $g(p)$ be the least primitive root modulo $p$ and $g^*(p)$ the least prime primitive root. Vinogradov conjectured that $g(p) \ll_\varepsilon p^\varepsilon$ for every $\varepsilon > 0$. Burgess [3] proved $g(p) \ll_\varepsilon p^{1/4+\varepsilon}$; for the prime version the least prime in a progression [9] gives $g^*(p) \ll p^{5.5}$. Under the generalised Riemann hypothesis Shoup [13] and Martin [10] proved $g^*(p) \ll \omega^4(\log\omega + 1)^4 \log^2 p$ with $\omega = \omega(p-1)$. Erdős [7] asked whether every prime $p$ has a prime $q < p$ that is a primitive root modulo $p$ (Problem 985 of [2]).

Let $\tau$ denote the divisor function. Throughout, $\chi$ is a Dirichlet character and $L(s,\chi)$ its $L$-function.

**Theorem 1.1.** *For every $\eta > 0$ there is $C_\eta$ such that every prime $p$ has a prime primitive root $\ell$ with*

$$\ell \le C_\eta\big(\tau(p-1)^2 \log(p+2)\big)^{8+\eta}.$$

**Corollary 1.2.** *For every $\varepsilon > 0$ there is $C_\varepsilon$ such that every prime $p$ has a prime primitive root $\ell \le C_\varepsilon p^\varepsilon$. In particular $g(p) \le g^*(p) \ll_\varepsilon p^\varepsilon$.*

**Corollary 1.3.** *There is $p_0$ such that every prime $p \ge p_0$ has a prime $q < p$ that is a primitive root modulo $p$.*

The input is the zero-free half-plane of [12], from which [12] deduces, as in [1, Thm. 6.7], that the least quadratic nonresidue is $\ll_\varepsilon p^\varepsilon$. A primitive root must avoid every subgroup of prime index at once, and the number of characters involved is the source of the factor $\tau(p-1)^2$.

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**Theorem 1.4** ([12])**.** *If $\chi$ is a nonprincipal Dirichlet character, then $L(s,\chi)\neq 0$ for $\operatorname{Re} s > 7/8$.*

Theorem 1.4 is formalised in Lean 4 in [12]. Theorem 1.1 and both corollaries are formalised here, with Theorem 1.4 as the only external input (Section 5). The constants are effective.

*Remark* 1.5. The constant $p_0$ of Corollary 1.3 is effective but far beyond direct computation. A computation (Section 5) gives $g^*(p) < p$ for every prime $3 \le p \le 10^8$, with $\max g^*(p) = 223$ at $p = 73189117$. Erdős's question therefore remains open for the primes in $[10^8, p_0)$.

*Remark* 1.6. Since $\tau(n) \le 2^{(1+o(1))\log n/\log\log n}$ [8, Thm. 317], Theorem 1.1 gives $g^*(p) \le \exp\bigl(c\log p/\log\log p\bigr)$ for an absolute constant $c$. This form is not formalised.

*Remark* 1.7. Carella has claimed $g^*(p) \ll p^{5/\log\log p}$ [4] and $g^*(p) = O((\log p)^{1+\varepsilon})$ [5, 6] without hypotheses. In [4], the estimate (16) bounds $\sum_{n\le x}\chi(n)\Lambda(n)n^{-2}$ by $O(1/x)$; the sum tends to $-L'(2,\chi)/L(2,\chi)$, which equals $-0.17264\ldots$ for the character modulo 3. In the arXiv version of [5], Lemma 7.1 asserts $E(x,y) \ll yp^{-\varepsilon}$ for the sum $E(x,y)$ of (7.2) when no integer of $[x, x+y]$ is a primitive root. Under that hypothesis every inner sum over $m$ equals $-1$, so $E(x,y) = -\varphi(p-1)\,\#([x,x+y]\cap\mathbb{Z})/p$ exactly, while $\varphi(p-1)/p \gg 1/\log\log p$; the step (7.2) replaces $\sum_n e(m\tau^n/p)$ by its value at $m=1$. We could not obtain the text of [6].

## 2. The prime sum of a nonprincipal character

For an arithmetic function $f$ and $x \ge 1$ put

$$S(f;x) = \sum_{n\le x} f(n)\Lambda(n)\Bigl(1-\frac{n}{x}\Bigr),$$

and write $S(\chi;x)$ when $f=\chi$. The aim of this section is Proposition 2.6.

**Lemma 2.1.** *Let $a\colon \mathbb{N}\to\mathbb{C}$ satisfy $\sum_n |a(n)|n^{-2} < \infty$ and put $F(s) = \sum_n a(n)n^{-s}$ for $\operatorname{Re} s \ge 2$. For $x > 0$,*

$$\sum_{n\le x} a(n)\Bigl(1-\frac{n}{x}\Bigr) = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(2+it)\,\frac{x^{2+it}}{(2+it)(3+it)}\,dt.$$

*Proof.* For $y>0$ let $k(y) = \frac{1}{2\pi i}\int_{(2)} y^s\,\frac{ds}{s(s+1)}$. If $y \ge 1$, moving the line to the left through the poles at 0 and $-1$ gives $k(y) = 1 - 1/y$; if $y<1$, moving it to the right gives $k(y)=0$. Since $|s(s+1)|^{-1}$ is integrable on $\operatorname{Re} s = 2$ and $\sum_n |a(n)|n^{-2} < \infty$, summation and integration may be exchanged: the right side equals $\sum_n a(n)k(x/n)$. □

**Lemma 2.2.** *Let $\chi$ be nonprincipal modulo $q$. For $\operatorname{Re} s = \sigma > 0$, $|L(s,\chi)| \le q|s|/\sigma$.*

*Proof.* Put $A(t)=\sum_{n\le t}\chi(n)$; then $|A(t)|\le q$, and partial summation gives $L(s,\chi)=s\int_1^\infty A(t)t^{-s-1}\,dt$ for $\sigma>0$. Hence $|L(s,\chi)|\le q|s|\int_1^\infty t^{-\sigma-1}dt = q|s|/\sigma$. □

**Lemma 2.3.** *Let $A_0=\frac{3}{2}\sum_\ell \ell^{-3/2}$, the sum over primes. For every Dirichlet character $\chi$ the series $E(s)=\sum_\ell -\log(1-\chi(\ell)\ell^{-s})$ is holomorphic on $\operatorname{Re}s>1$, $e^{E(s)}=L(s,\chi)$ there, and $|E(s)|\le A_0$ for $\operatorname{Re}s\ge 3/2$.*

*Proof.* For $\operatorname{Re}s\ge 1$ we have $|\chi(\ell)\ell^{-s}|\le 1/2$, and $|\log(1-w)|\le\frac{3}{2}|w|$ for $|w|\le 1/2$. The series therefore converges locally uniformly on $\operatorname{Re}s>1$ and is bounded by $A_0$ on $\operatorname{Re}s\ge 3/2$, and the Euler product gives $e^E=L$. □

**Lemma 2.4.** *Let $0<\delta<1/1000$, $1/2\le a\le 1$, and let $\chi$ be nonprincipal modulo $q$ with $L(s,\chi)\ne 0$ for $\operatorname{Re}s>a+2\delta$. Put $K_1=(1+2A_0)/(2\delta^2)$. For $a+8\delta\le\sigma\le 2$ and $t\in\mathbb{R}$,*

$$\left|\frac{L'}{L}(\sigma+it,\chi)\right|\le K_1\log\bigl(2q(3+|t|)^2\bigr).$$

*Proof.* Let $c=2+it$, $R=2-a-3\delta$, $r=2-a-6\delta$ and $\Lambda_0=\log(2q(3+|t|)^2)\ge\log 18>1$. On the disc $|z-c|<2-a-2\delta$ we have $\operatorname{Re}z>a+2\delta$, so $L(z,\chi)$ has no zero there and has a holomorphic logarithm $h$. On $|z-c|<1/2$ both $h$ and $E$ of Lemma 2.3 are logarithms of $L$, so they differ by a constant in $2\pi i\mathbb{Z}$; we choose $h=E$ there. Then $|h(c)|\le A_0$.

If $|z-c|\le R$, then $\operatorname{Re}z\ge a+3\delta\ge 1/2$ and $|z|\le|c|+3/2\le 4+|t|$, so Lemma 2.2 gives $\operatorname{Re}h(z)=\log|L(z,\chi)|\le\log(2q(4+|t|))\le\Lambda_0$. The Borel–Carathéodory inequality [14, §5.5] gives, for $|z-c|\le r$,

$$|h(z)-h(c)|\le\frac{2r}{R-r}\max_{|w-c|=R}\operatorname{Re}\bigl(h(w)-h(c)\bigr)\le\frac{\Lambda_0+A_0}{\delta},$$

as $r\le 3/2$ and $R-r=3\delta$. Hence $|h(z)|\le A_0+(\Lambda_0+A_0)/\delta\le(1+2A_0)\Lambda_0/\delta$ on $|z-c|\le r$. For $w=\sigma+it$ with $a+8\delta\le\sigma\le 2$ we have $|w-c|=2-\sigma\le r-2\delta$, so the circle $|z-w|=2\delta$ lies in $|z-c|\le r$, and Cauchy's estimate gives $|L'/L(w,\chi)|=|h'(w)|\le(1+2A_0)\Lambda_0/(2\delta^2)$. □

**Lemma 2.5.** *For $q\ge 1$, $t\in\mathbb{R}$ and $s=\sigma+it$ with $\sigma\ge 1/2$,*

$$\log\bigl(2q(3+|t|)^2\bigr)\le 10\log(q+2)(1+|t|)^{1/2},\qquad |s(s+1)|\ge\tfrac{1}{8}(1+|t|)^2.$$

*Proof.* We have $\log(2q)\le 2\log(q+2)$, and $\log y\le 2y^{1/2}$ with $3+|t|\le 4(1+|t|)$ gives $\log(3+|t|)\le 4(1+|t|)^{1/2}$; the first bound follows from $\log(q+2)\ge 1$. For the second, $|s|\ge\max(\sigma,|t|)\ge(1+|t|)/4$ and $|s+1|\ge\max(\sigma+1,|t|)\ge(1+|t|)/2$. □

**Proposition 2.6.** *Let $0<\delta<1/1000$. There is $K=K(\delta)$ such that the following holds. Let $1/2\le a\le 1$ and let $\chi$ be nonprincipal modulo $q$ with $L(s,\chi)\ne 0$ for $\operatorname{Re}s>a+2\delta$. Then for $x\ge 1$,*

$$|S(\chi;x)|\le K\,x^{a+8\delta}\log(q+2).$$

*Proof.* Put $\sigma_0 = a + 8\delta$ and

$$\Phi(s) = -\frac{L'}{L}(s,\chi)\,\frac{x^s}{s(s+1)}.$$

For $\operatorname{Re} s > 1$, $-L'/L(s,\chi) = \sum_n \chi(n)\Lambda(n)n^{-s}$, so Lemma 2.1 gives $S(\chi;x) = \frac{1}{2\pi}\int \Phi(2+it)\,dt$. The function $\Phi$ is holomorphic on $\sigma_0 \le \operatorname{Re} s \le 2$, and by Lemmas 2.4 and 2.5,

$$|\Phi(\sigma+it)| \le 80K_1 \log(q+2)\, x^{\sigma}(1+|t|)^{-3/2} \qquad (\sigma_0 \le \sigma \le 2).$$

This tends to 0 as $|t| \to \infty$ uniformly in $\sigma$, so Cauchy's theorem on the rectangles $[\sigma_0, 2] \times [-T, T]$, $T \to \infty$, moves the integral to $\operatorname{Re} s = \sigma_0$. Since $\int_{\mathbb{R}} (1+|t|)^{-3/2} dt = 4$, we get $|S(\chi;x)| \le \frac{160K_1}{\pi} x^{\sigma_0} \log(q+2)$. □

## 3. Primitive roots as a combination of characters

Fix a prime $p$, put $N = p-1$, let $g$ generate $(\mathbb{Z}/p\mathbb{Z})^\times$, and let $\chi_1$ be the character modulo $p$ with $\chi_1(g) = e^{2\pi i/N}$. Let $\chi_0$ be the principal character.

**Lemma 3.1.** *For every integer $n$,*

$$\mathbf{1}[n \text{ is a primitive root mod } p] = \sum_{d \mid N} \frac{\mu(d)}{d} \sum_{m=0}^{d-1} \chi_1^{mN/d}(n),$$

*and $\chi_1^{mN/d} \ne \chi_0$ for $d \mid N$ and $0 < m < d$.*

*Proof.* If $p \mid n$ both sides vanish. Otherwise $n \equiv g^k$, the order of $n$ is $N/(N,k)$, and $\sum_{m<d} e^{2\pi i k m/d} = d\,\mathbf{1}[d \mid k]$; the right side is $\sum_{d \mid (N,k)} \mu(d) = \mathbf{1}[(N,k) = 1]$. The character $\chi_1^{mN/d}$ sends $g$ to $e^{2\pi i m/d} \ne 1$. □

**Lemma 3.2.** *For $N \ge 1$, $\sum_{d \mid N} \mu(d)/d = \varphi(N)/N \ge 1/\tau(N)$.*

*Proof.* $\varphi(N)/N = \prod_{r \mid N}(1 - 1/r) \ge 2^{-\omega(N)} \ge 1/\tau(N)$, the product over the prime divisors $r$ of $N$. □

**Lemma 3.3.** *For $x \ge 10^4$, $\operatorname{Re} S(\chi_0;x) \ge \frac{1}{2}\psi(x/2) - \log x \ge x/10$.*

*Proof.* The terms of $\sum_{n \le x} \Lambda(n)(1 - n/x)$ are nonnegative and at least $\Lambda(n)/2$ for $n \le x/2$, so the sum is at least $\psi(x/2)/2$. The terms removed by $\chi_0$ are those with $p \mid n$, that is $n = p^k \le x$, and they contribute at most $\sum_{p^k \le x} \log p \le \log x$. For the second inequality, $\psi(n) \ge n \log 2$ for integers $n \ge 7$ [11], so $\psi(x/2) \ge (x/2 - 1)\log 2$, and $\frac{x}{4}\log 2 - \frac{1}{2}\log 2 - \log x \ge \frac{x}{10}$ for $x \ge 10^4$. □

**Lemma 3.4.** *If $\ell^k$ with $k \ge 1$ is a primitive root modulo $p$, so is $\ell$.*

*Proof.* $\ell$ is a unit modulo $p$, and $N = \operatorname{ord}(\ell^k)$ divides $\operatorname{ord}(\ell)$, which divides $N$. □

## 4. Proofs of the theorem and the corollaries

*Proof of Theorem 1.1.* Let $\delta = \eta/(2000(8+\eta))$, so $0 < \delta < 1/1000$, and let $K$ be given by Proposition 2.6; put $K' = \max(K, 1)$. Let $\sigma_0 = 7/8 + 8\delta$ and $\beta = 1 - \sigma_0 = 1/8 - 8\delta > 0$. Then

$$(8+\eta)\beta = 1 + \frac{\eta}{8} - \frac{\eta}{250} \geq 1.$$

Put $C_\eta = \max\bigl(10^4, (20K')^{8+\eta}\bigr)$. Given a prime $p$, let $\tau = \tau(p-1)$, $Y = \tau^2 \log(p+2) \geq 1$ and $x = C_\eta Y^{8+\eta} \geq 10^4$.

By Theorem 1.4, Proposition 2.6 applies with $a = 7/8$ to every nonprincipal $\chi$ modulo $p$, so $|S(\chi; x)| \leq B := K' x^{\sigma_0} \log(p+2)$. Let $G$ be the sum $S(f; x)$ with $f$ the indicator of the primitive roots modulo $p$. By Lemma 3.1,

$$G = c_0 S(\chi_0; x) + \mathcal{E}, \qquad c_0 = \sum_{d \mid N} \frac{\mu(d)}{d}, \qquad \mathcal{E} = \sum_{d \mid N} \frac{\mu(d)}{d} \sum_{0 < m < d} S(\chi_1^{mN/d}; x),$$

and $|\mathcal{E}| \leq \sum_{d \mid N} \frac{d-1}{d} B \leq \tau B$. Lemmas 3.2 and 3.3 give

$$|G| \geq c_0 \operatorname{Re} S(\chi_0; x) - |\mathcal{E}| \geq \frac{x}{10\tau} - \tau B.$$

Now $x^\beta = C_\eta^\beta Y^{(8+\eta)\beta} \geq (20K')^{(8+\eta)\beta} Y \geq 20K'Y$, so $\tau^2 B = K' Y x^{\sigma_0} \leq x^{\beta+\sigma_0}/20 = x/20$. Hence $|G| \geq x/(20\tau) > 0$, and some $n \leq x$ with $\Lambda(n) \neq 0$ is a primitive root modulo $p$. Writing $n = \ell^k$, the prime $\ell \leq x$ is a primitive root by Lemma 3.4. □

*Proof of Corollary 1.2.* Take $\eta = 1$ and $\nu = \varepsilon/27$. By [8, Thm. 315] there is $D_\nu$ with $\tau(n) \leq D_\nu n^\nu$, and $\log(p+2) \leq (p+2)^\nu/\nu \leq 3^\nu p^\nu/\nu$. Theorem 1.1 gives $\ell \leq C_1 (D_\nu^2 3^\nu \nu^{-1})^9 p^{27\nu}$. □

*Proof of Corollary 1.3.* Corollary 1.2 with $\varepsilon = 1/2$ gives $\ell \leq C\sqrt{p} < p$ for $p > C^2$. □

## 5. Formal verification and computations

The Lean 4 development (Lean v4.34.1, Mathlib v4.34.1) proves

```
theorem least_prime_primitive_root_bound {η : ℝ} (hη : 0 < η) :
    ∃ C : ℝ, 0 < C ∧ ∀ p : ℕ, p.Prime → ∃ ℓ : ℕ, ℓ.Prime ∧
      orderOf (ℓ : ZMod p) = p - 1 ∧
      (ℓ : ℝ) ≤ C * (((p - 1).divisors.card : ℝ) ^ 2 *
        Real.log (p + 2)) ^ (8 + η)
theorem least_prime_primitive_root_le_rpow {ε : ℝ} (hε : 0 < ε) :
    ∃ C : ℝ, 0 < C ∧ ∀ p : ℕ, p.Prime → ∃ ℓ : ℕ, ℓ.Prime ∧
      orderOf (ℓ : ZMod p) = p - 1 ∧ (ℓ : ℝ) ≤ C * (p : ℝ) ^ ε
theorem erdos_985_of_large : ∃ p₀ : ℕ, ∀ p : ℕ, p.Prime →
    p₀ ≤ p → ∃ q : ℕ, q.Prime ∧ q < p ∧
      orderOf (q : ZMod p) = p - 1
```

which are Theorem 1.1 and Corollaries 1.2 and 1.3. Theorem 1.4 enters as the theorem

`OAI.DirichletCharacter.LFunction_ne_zero_of_seven_eighths_lt_re`

of [12], stated for Mathlib's `DirichletCharacter.LFunction` and compiled from its source. The Lean proof of Lemma 2.4 uses the logarithmic control lemma of [12] in place of the Borel–Carathéodory argument, and the proof of Lemma 3.3 uses Mathlib's bound $\psi(y) \geq (y-1)\log 2 - \log(y+2)$ in place of [11]. Every theorem depends only on the axioms `propext`, `Classical.choice` and `Quot.sound`.

Programs in Python, C, C++ and Julia compute $g^*(p)$ for all primes $3 \leq p \leq 10^8$ (Remark 1.5), check Lemma 3.1 numerically for $p \leq 200$, the inequality of Lemma 3.3 on $10^4 \leq x \leq 2 \cdot 10^7$, Lemma 3.2 for $N \leq 10^7$, the exponent identity in the proof of Theorem 1.1 in exact arithmetic, and the two values in Remark 1.7.

## Data availability

The Lean project, the programs and a script that runs them are archived at Zenodo, doi:10.5281/zenodo.0000000.

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